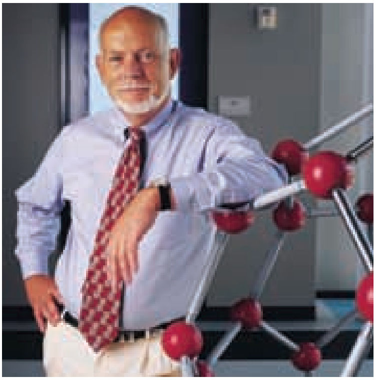
Richard E. Smalley

Yet imagine if this nanobot was so versatile that it could build anything, as long as it had a supply of the right kinds of atoms, a source of energy and a set of instructions for exactly what to build. If the nanobot could really build anything, it could certainly build another copy of itself. It could therefore self-replicate, much as biological cells do. After a while, we'd have a second nanobot and, after a little more time, four, then eight, then 16 and so on.

Suppose each nanobot consisted of a billion atoms (109 atoms) in some incredibly elaborate structure. If these nanobots could be assembled at the full billion-atoms-per-second rate imagined earlier, it would take only one second for each nanobot to make a copy of itself. The new nanobot clone would then be "turned on" so that it could start its own reproduction. After 60 seconds of this furious cloning, there would be 260 nanobots, which is the incredibly large number of 1 × 10^18, or a billion billion. This massive army of nanobots would produce 30 grams of a product in 0.6 millisecond, or 50 kilograms per second. Now we're talking about something very big indeed!

Although it is a very pretty picture, Smalley asks us to consider how realistic this notion of self-replicating nanobots is. Atoms are tiny and move in a defined and circumscribed way — so as to minimize the free energy of their local surroundings. The electronic "glue" that sticks them to one another is not local to each bond but rather is sensitive to the exact position and identity of all the atoms in the near vicinity. So when the nanomanipulator arm of our nanobot picks up an atom and goes to insert it in the desired place, it has a fundamental problem. It also has to somehow control not only this new atom but all the existing atoms in the region. Even if the nanobot has an additional manipulator arm for each one of these atoms, our problems don't end here. The region where the chemistry is to be controlled by the nanobot is very, very small—about one nano on a side.

This constraint led Smalley to raise two fundamental difficulties or problem

- The fat fingers problem.
- The sticky fingers problem.

Because the fingers of a manipulator arm must themselves be made out of atoms, they have a certain irreducible size. There just isn't enough room in the